Metric

DOD-P-63252 (AR)
29 January 1979

PROJECTILE, 155MM, HE, XM795 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing of the 155MM, HE, XM795 Projectile.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-453 - Inspection, Radiographic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1320

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DRAWINGS

ARRADCOM

9312769 -	Projectile, 155MM, HE, XM795 Loading, Assy & Marking
9329574 -	Packing and Marking for Pallet Wood Ammunition for Projectile, 155MM, HE, XM795

3. REQUIREMENTS

3.1 Materials.-Materials and parts shall be in accordance with the applicable drawings and specifications.

3.1.1 Clean explosive scrap.-Clean explosive scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scraps from floor sweepings, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.5).

3.2 Assembled projectile.-The assembled projectile shall comply with all requirements specified on Drawing (Dwg.) 9312769, and with all requirements specified in the applicable specifications.

3.3 Cavitation in high explosive charge

3.3.1 Cracks and projected cavities.-Cracks and cavities within the explosive charge shall not exceed the requirements specified in Table I when determined in accordance with 4.4.3.1. Cavities having a maximum (max) projected length of 0.8mm or less shall be disregarded, except in Segment A. Any defect in Segment A that is of sufficient size to register an indication on the radiograph shall be cause for rejection regardless of its projected area.

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TABLE I

	SEGMENTS (See Figure 1)			
	A	B	C	D
Sum of projected areas of the cavities, excluding pipes, cracks and annular rings, square millimeters (sq. mm)	0	161.0	323.0	323.0
Projected length of any cavity excluding pipes, cracks, and annular rings (mm).	0	12.7	12.7	19.0
Piping cavities max. projected area (sq. mm).	0	161.0	323.0	-
Piping cavities max. projected width (mm).	0	6.4	6.4	-
Cracks, max. projected width (mm)	0	0.8	0.8	-
Annular rings, max. projected width (mm)	0	0	6.4	-

Piping cavitation is defined as cavitation located on or near the vertical center line of the projectile with the longitudinal axis (length) three times or greater than the width.

An annular ring is defined as cavitation that takes the form of a ring or portion thereof around the periphery of the cast.

3.3.2 Surface cavitation.-Cavities formed due to chipped or broken explosive from the sidewalls of the fuze well shall not extend in aggregate around more than 1/4 of the circumference. The max. dimension of any visible cavity in the faced surface of the charge shall not exceed 6.4mm in any direction and the total area of all cavities shall not exceed 80 percent of the surface area. (Cavities having a max. dimension of 0.8mm or less shall be disregarded in this computation as shall voids within the applicable forward region of the fuze well prescribed by dwg. 9312769).

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3.3.3 Porous areas.-Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.4 Cracked charges.-No transverse crack shall be permitted in Segment A and not more than a total of two transverse cracks shall be permitted in Segments B and C combined. Cracks shall be disregarded in Segment D.

3.3.5 Base separation.-There shall be no discernible base separation between the base of the projectile cavity and the explosive charge when inspected by film radiography. The maximum allowable base separation shall be .015 inches when determined by sectioning.

3.4 Loading of high explosive

3.4.1 Mating surfaces.-The mating surfaces of threads shall be free of caked explosive and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care shall be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the projectile.

3.4.2 Solid explosives.-Solid explosive shall not be in the projectile before loading or added to the projectile during initial pouring or cooling.

3.4.3 Coating.-Projectile having the interior wall imperfectly coated shall not be loaded. The interior surface of the projectile shall be dry when the high explosive is poured into the projectile (see 4.4.2.1).

3.4.4 Projectile interior cavity.-The interior cavity of the projectile prior to loading shall be free from foreign matter.

3.4.5 Fuze well.-The fuze well shall be free from any foreign matter or loose explosive prior to assembly of the liner.

3.4.6 Unsolidified pool of paint.-There shall be no unsolidified pool of paint or blister in the projectile cavity prior to loading of the explosive.

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3.4.7 Exterior surfaces.-There shall be no explosive on the exterior surface of the projectile after loading.

3.4.8 Body cavity.-The body cavity shall be free of scale, fins, burrs, draw marks, laminations, embedded foreign matter, folds, pits and sharp edges.

3.5 Proving ground.-There shall be no premature burst or indication of metal parts separation in the gun bore or in flight.

3.6 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Exudation.-After heat cycling of HE filled projectiles, there shall be no exudation in the fuze well or exterior of the projectile.

3.8 Workmanship.-The requirements of workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.8.1 Cleaning.-If required the cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3.8.2 Protection of the rotating band.-Care shall be exercised at all times so that the rotating band of each projectile will not be nicked or burred.

3.8.3 Painting.-Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.8.4 Foreign matter.-No part or assembly shall contain dirt, grease, chips, rust corrosion or other foreign matter, prior to inspection and shipping.

3.8.5 Obturator.-Care shall be taken at all times to protect obturator from being damaged.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspections shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Projectile	9312769	24

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63252 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Projectile, 155mm, XM795				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, 155mm, HE, XM795				
	Loading, Assy and Marking (Dwg. 9312769)				
	Examination for Defects	16		3.2	4.4.2.1 to 4.4.2.5
	Proving Ground Test	16		3.5	4.5.2
	Base Separation	8		3.3.5	4.4.3.1.1
	Special Radiographic inspection of sectioned projectile. After the test for base separation, each sample shall be completely sectioned and each half radiographed for the purpose of detailed examination of the explosive filler.			3.3	4.5.1.2 to 4.5.1.6, except that any convenient orientation may be used for radiography of sectioned projectiles.

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 QUALITY CONFORMANCE INSPECTION

4.4.1 Inspection lot formation.-Inspection lots shall conform with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of loaded projectiles shall contain:

a. Projectile metal parts of one lot interfix number from one manufacturer.

b. Loaded projectile of one weight zone.

c. Fuze well liners from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification, and same specification revision.

d. TNT of the same type from one lot interfix number from one manufacturer.

4.4.2 Examinations.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63252 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Projectile Prior to Loading				9312769
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Foreign matter in projectile cavity		100%	3.4.4	Visual
2	Unsolidified pool of paint or blister in projectile cavity		100%	3.4.6	Visual/Manual
3	Interior metal defects		100%	3.4.8	Visual
Major					
101	Interior protective coating with bare spot		0.40%	3.4.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63252 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Projectile and Liner (Prior to Assembly of Liner)				9312769
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth of fuze well, max.		0.40%	3.2	Gage
102	Foreign matter in fuze well		0.40%	3.4.5	Visual
Minor					
201	Silicone compound not 360 degrees around liner or on its base (exterior surfaces only)		100%	3.2	Visual
202	Diameter of fuze well, maximum		0.65%	3.2	Gage
203	Surface cavitation excessive		0.65%	3.3.2	Visual
204	Evidence of poor workmanship		0.65%	3.8	Visual
205	Presence of explosive on exterior of shell body		100%	3.4.7	Visual

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CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63252 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Projectile, Prior to Assembly of Lifting Plug				9312769
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Liner missing		100%	3.2	Visual
Major					
101	True position of inside diameter of liner with pitch diameter of thread		0.40%	3.2	Gage
102	Depth to bottom of liner, min.		0.40%	3.2	Gage
Minor					
201	Liner damaged		0.65%	3.2	Visual
202	Foreign matter in liner cavity including explosive		0.65%	3.2	Visual
203	Silicone compound missing from threads		0.65%	3.2	Visual
204	Explosive in thread		0.65%	3.4.1	Visual
205	Evidence of poor workmanship		0.65%	3.8	Visual
206	Liner removable by hand		0.65%	3.2	Manual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63252 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Projectile Assembly		9312769
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Projectile zone weight incorrect		100%	3.2	Balance
2	Zone stenciling incorrect		100%	3.2	Visual
Major					
101	Diameter of bourrelet, max. (applies if bourrelet has been retouched)		100%	3.2	Gage
102	Torque test failure of lifting plug		0.40%	3.2	Test
103	Lifting plug gasket missing		0.40%	3.2	Visual
104	Any weight zone punch mark inadequate		0.40%	3.2	Visual/Manual
Minor					
201	Grommet missing, damaged or improperly assembled		0.65%	3.2	Visual
202	Rotating band damaged		0.65%	3.8.2	Visual
203	Protective coating with total damaged area in excess of 161 square mm		0.65%	3.8.3	Visual
204	Marking misleading or unidentifiable (except for zones), including incorrect color code		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.8	Visual
206	Obturator missing or damaged		100%	3.8.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63252 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Pallet				9329574
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pallet cover damaged or improperly assembled		0.40%	3.2	Visual
102	DOD symbol missing or unidentifiable		0.40%	3.2	Visual
103	Projectile of different zones on the same pallet		0.40%	3.2	Visual
104	Strapping or wire missing or loose		0.40%	3.2	Manual
Minor					
201	Number of projectile assemblies incorrect		0.65%	3.2	Visual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Strapping or wire improperly engaged		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Cavitation (see 3.3 except 3.3.2).-Cavitation defects are classified as follows:

Critical:

1. Any cavitation, porosity, cracks, piping, or annular rings in Segment A.....Radiography
2. Cavitation, porosity, cracks, piping or annular rings in Segment B or C in excess of that permitted in Table I.....Radiography
3. Cracks in excess of that permitted in 3.3.4.....Radiography
4. Base separation (see 4.4.3.1.1).....Radiography
5. Cavitation in Segment D in excess of that permitted in Table I.....Radiography

This test shall be performed 100 percent on all projectile assemblies. Each assembly shall be subject to radiography examination. The test shall be performed as specified in 4.5.1 using equipment in accordance with 4.4.4. Any assembly which has any of the above critical defects shall be classed defective and removed from the lot.

4.4.3.1.1 Base separation.-Select projectiles at the rate of one (1) out of seven hundred (700) projectiles to be included in the post cycle heat treatment lot with a minimum of two (2) samples per lot. These projectiles are to be included in the post cycle heat treatment and sectioned for determination of base separation. The samples selected for sectioning must be placed in the coolest area of the post cycle igloo. Any projectile after sectioning which has base separation in excess of .015 inches shall be classed defective, and the entire post cycle heat treatment lot rejected.

4.4.3.2 Proving ground.-The test shall be conducted in accordance with 4.5.2. If any projectile fails to comply with the applicable requirement the lot shall be rejected.

4.4.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with all the acceptance criteria contained herein 16 sample projectiles shall be selected at random from each lot for this test. If any critical defect occurs, the lot shall be rejected.

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4.4.3.2.2 After three consecutive lots have complied with 4.4.3.2.1, 8 sample projectiles shall be selected at random for this test. If any critical defect occurs, the lot shall be rejected. In the event that two successive lots fail to meet all the requirements of this paragraph, the provisions of 4.4.3.2.1 shall again apply until three successive lots meet the requirements specified therein.

4.4.3.2.3 Test validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be considered invalid and a new test shall be performed with additional samples.

4.4.3.3 Exudation - Minor defect.-Eight projectiles from the hottest area of the post cycle oven shall be used for the sample. Any projectile with exudation shall be cause for rejection of the post cycle treatment lot. Inspection shall be performed per 4.5.3.

4.4.4 Inspection equipment.-Requirements for inspection equipment to perform examinations and tests are specified in the item detail specification and in specific contract clauses as applicable. See Section 6 of this specification for general provisions applicable to inspection equipment designs.

4.5 Test Methods and Procedures

4.5.1 Radiographic examination

4.5.1.1 Projectile preparation.-Immediately preceding radiographic examination the projectile shall be subjected to an acceleration level of 200 to 250 G's in the nose down orientation (see Note 6.9) forcing explosive cast toward nose. Following this step the projectile shall not be turned onto its base until radiographic examination has been completed.

4.5.1.2 Determination of projectile cavitation.-All projectiles shall be subjected to radiographic examination in the nose down orientation. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the projectile and tangent to the base of its internal cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks,

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or other extraneous features should project into the region of explosive under inspection, with the exception of specified penetrameters and radiation screening required to reduce film densities to readable levels in the thinner areas of the shell. Such screening should be free of any radiographically observable features in the projected area under inspection. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. In addition, the sensitivity of the radiographic technique shall be 2.0 percent and the minimum photographic density of the film used shall be no less than 1.25 at every point within the projected area of explosive, including the base, with the maximum density no greater than to permit adequate inspection over the entire area of interest, as determined in accordance with 4.5.1.3 and 4.5.1.4. All defect-like features in the area of interest on a radiographic image shall be treated as defects unless identified as extraneous artifacts or metal parts irregularities. In the event an original radiograph leaves doubt as to a shell's acceptability because of difficulty in evaluating the size, location, or classification of a defect, the projectile shall be rotated about its longitudinal axis from the original exposure, and additional radiographs taken for clarification. Acceptability in all cases shall be judged on the basis of the film giving the largest projection, except in the case where a re-radiograph reveals multiple superimposed cavities, each of which individually complies with the requirements. All additional radiographs of a projectile for defect conditions will be identified on the film as retakes and stored; traceable to the original.

4.5.1.3 Check of sensitivity of radiographic technique.-The sensitivity of radiographic technique shall be determined by means of penetrameters fabricated as specified herein and through the use of standards containing known defects. The penetrameters shall be placed on the radiographic source side of a projectile so that the 2T holes are at the point shown in Figure 2. Three penetrameters for each projectile shall be used. When radiographing a number of similar projectiles the frequency of the sensitivity check shall be no less than one in each fifty radiographs, and on a rotating ring, the penetrameter shall be used at one station for each complete rotation of the ring. The use of standards will be limited to the initial qualification of the radiographic technique at the start of production and at intervals stated in 4.5.1.5 provided that uniform results can be expected during the interval.

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The sensitivity of the radiographic technique shall be considered satisfactory when the image of the penetrameters, including the 2T holes, are perceptible in accordance with MIL-STD-453 and when the required defect standards are perceptible.

The penetrameters shall be made of steel having an approximate specific gravity of 7.8. The penetrameters shall be located as shown on Figure 2. For further details on construction of the penetrameter see MIL-STD-453. The standard projectile will be provided by ARRADCOM.

4.5.1.4 Check of photographic density of film.-The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units. Accuracy of densitometer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurement on the radiographic film. Calibration shall be made at a step tablet density level within .5D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) or no more than 6mm in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by ARRADCOM. Reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.5.1.5 Review of radiographic sensitivity and techniques (see 6.7).-The sensitivity of the radiographic technique shall be reviewed by submitting at least one radiograph of the standard and of the projectile containing the penetrameter and a copy of radiographic procedures prior to the start of initial production. These radiographs shall be taken and processed

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using the same techniques as will be used in production. In addition one radiograph of the projectile standard and the penetrameter projectile shall be submitted for review once a month for the first three months of each contract and every three months thereafter. After qualification of the radiographic technique no changes to developing process, film, manufacturer or type radiographic source or exposure time shall be made without first submitting additional sample radiographs of the standard and penetrameter projectile for approval.

4.5.1.6 Radiographic film.-Only a film type and vendor as listed herein has been tested and approved by ARRADCOM for use in radiographing this projectile. The final choice of film type must be determined for the particular radiographic facility to be used and will be based on achieving adequate sensitivity on both standard and penetrameter projectile. A substitute item shall not be used before prior testing and/or approval by ARRADCOM, Dover, New Jersey 07801.

Eastman Kodak Company
1187 Ridge Road, West
Rochester, New York 14650

Film Types: T or M

E. I. DuPont de Nemours and Company
Photo Products Department
Wilmington, Delaware 19898

Film Types: NDT-55

GAF Corporation
140 West 51 Street
New York, New York 10020

Film Types: 400

Processing chemicals and conditions shall be as recommended by film manufacturer.

4.5.2 Proving ground test.-This test shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The samples selected in accordance with 4.4.3.2 and Table II shall be fired in a M199 series cannon. The sample projectiles shall be assembled with a dummy or inert loaded fuze and thereafter divided into two equal groups. Group 1 shall be conditioned and fired at -54°

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Celsius at Charge 8, M203 series propelling charge conditioned to -54°C. Group 2 shall be conditioned and fired at 63°C with a Charge 8 M203 propelling charge conditioned to 63°C. Radar observation shall be made for muzzle velocity and range and deflection of impact (see 6.6.1). The test results are to be recorded for individual items. Camera coverage shall be provided to observe the projectile in flight in order to record any malfunctions (see 6.6). In addition, radar shall be used to observe the projectile in flight in order to record any possible malfunctions.

4.5.3 Exudation.-The exterior of the projectile shall be visually inspected for exudation. If no exudate is found remove the lifting plug and inspect for exudation visually.

4.5.4 Determination of base separation.-After heat treatment the samples from each post cycle heat treatment lot shall be cooled to ambient temperature. The projectiles shall then be conditioned for a minimum of eight (8) hours at 21 ± 5°C and sectioned at 21 ± 5°C. A notch shall be cut from the base in the following manner: Cut parallel to the base approximately 90mm up from the base to the center of the projectile axis, next cut through the base toward the nose perpendicular to aforementioned cut so as to remove a wedge of steel to expose explosive and metal parts interface. Next subject projectile to an acceleration level of 200 to 250 G's in the nose down orientation (see 4.5.1.1) and maintain projectile in that orientation to measure for base separation with a 1.25cm wide feeler gage.

5. PREPARATION FOR DELIVERY

5.1 Packing - Level A.-The projectile shall be packed in accordance with dwg. 9329574.

5.2 Marking.-Marking shall be in accordance with dwg. 9327883.

6. NOTES

6.1 Intended use.-The components and assemblies covered by this specification are intended for use on the XM795 HE Projectile.

6.2 Ordering data.-See MIL-A-48078.

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6.3 Submission of inspection designs for approval - See MIL-A-48078.-Submit equipment designs as required to Commander, ARRADCOM, Attn: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Photographic densities.-"American Standards for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, has been satisfactory as a source of information on film densities.

6.5 Clean explosive scrap.-Clean explosive scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations and scrap formed during the drilling operation.

6.6 Proving ground.-A copy of all proving ground data should be submitted to Commander, US Army Armament Research and Development Command, Attn: DRDAR-QAR-Q and DRDAR-LCU-, Dover, New Jersey 07801.

6.6.1 Range and deflection.-Range and deflection are not technical requirements of this specification. They are mandatory performance requirements that are achieved by design and are affected by variability in the manufacturing process that may not be completely controlled by the present TDP.

6.7 Radiographic procedures.-Radiographic procedures, and sample radiographs shall be submitted to the Commander, ARRADCOM, Dover, New Jersey 07801, Attn: DRDAR-QAR-Q for review and approval.

6.8 Visual examination qualification.-When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.9 Projectile G loads.-The 200 to 250 G load acceleration is applied to simulate projectile ramming. This is based on M107/M110 self-propelled artillery loader-rammer review program dated June 1971.

Custodian:
Army - AR

Preparing activity:
Army - AR

Project No. 1320-A414

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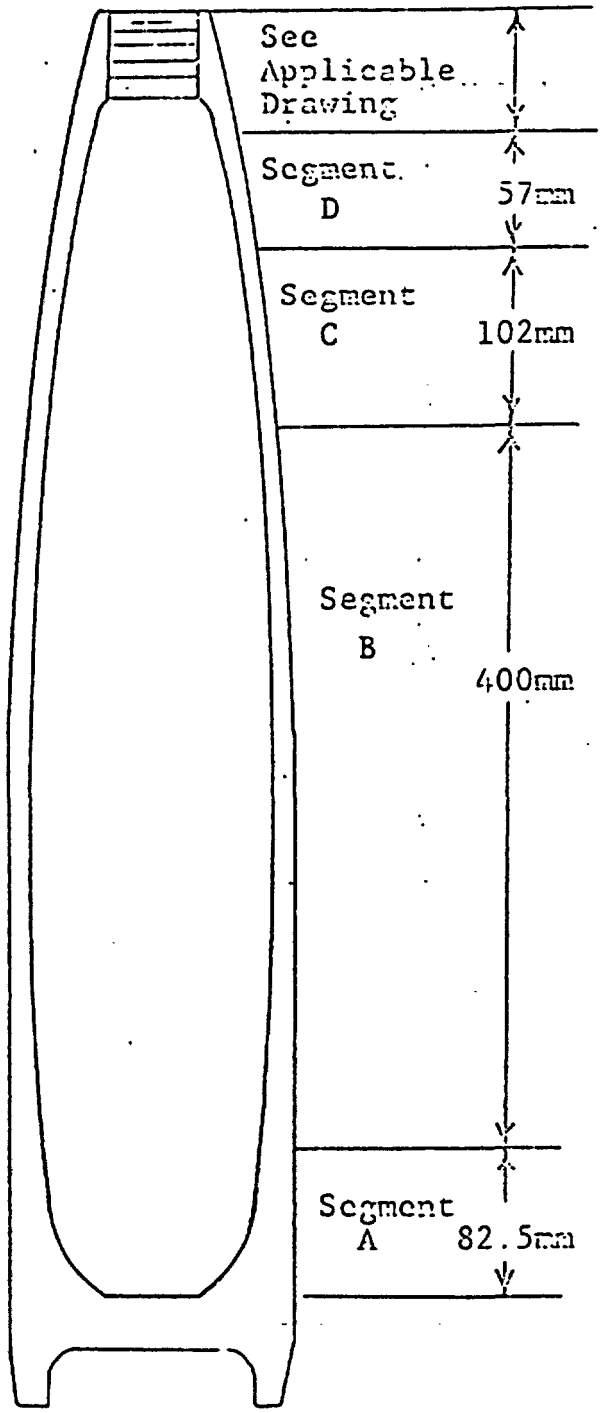
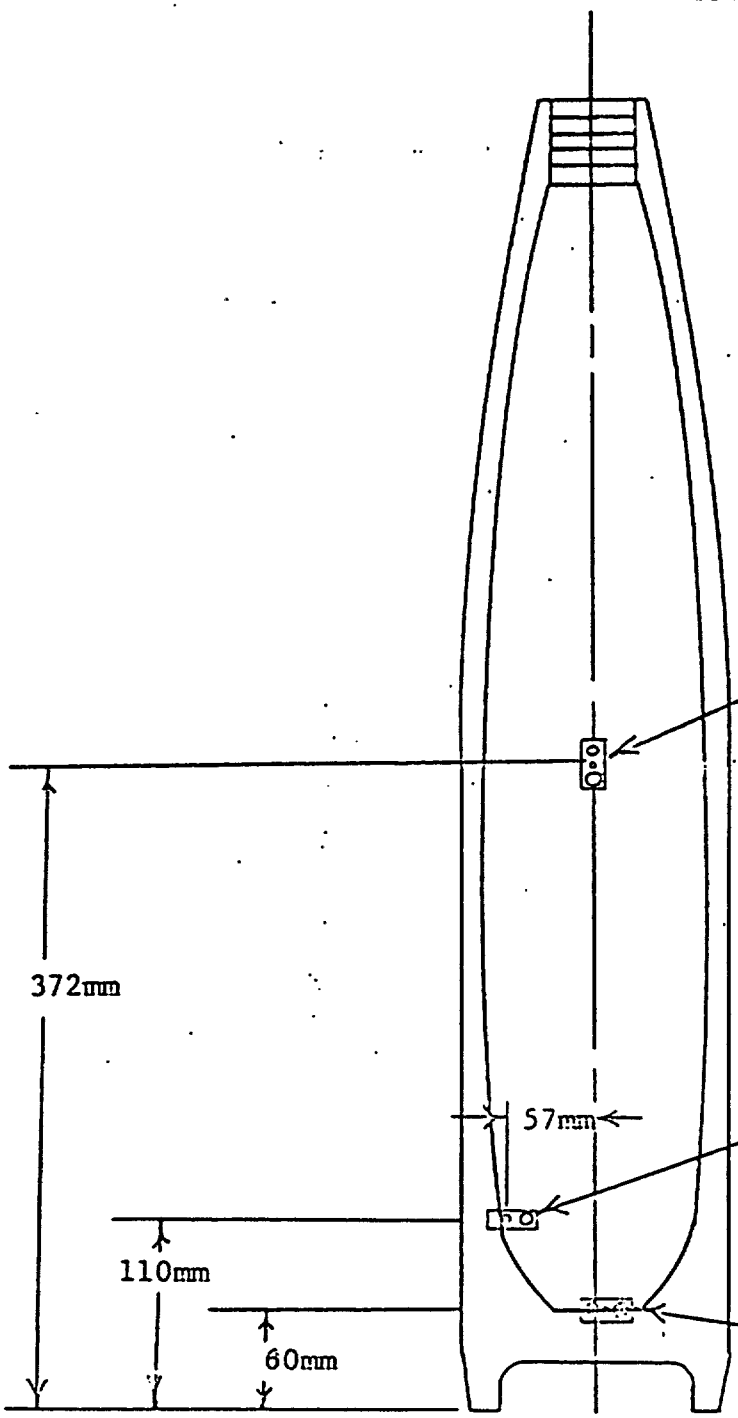


FIGURE 1

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Penetrameter constructed per MIL-STD-453
Dimensions shown are for location of center of 2T hole



12.5mm single steel wall
130mm explosive penetrameter thickness
1.0mm
2T hole dia = 2.0mm

49mm single steel wall
181mm explosive penetrameter thickness
2.0mm
2T hole dia = 4.0mm

53mm single steel wall
49mm explosive penetrameter thickness
2.3mm
2T hole dia = 4.6mm
position 2T hole so as tangent to the cavity base and on the central axis of projectile

Projectile, 155mm, HE, XM795
Radiographic Penetrameter Requirements

FIGURE 2

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

DOD-P-63252 (AR) Projectile, 155MM, HE, XM795 Loading, Assembling and Packing

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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